

New York City Airbnb Dataset Report

1. Dataset Structure

1.1 Overview

The New York City Airbnb dataset (AB_NYC_2019.csv) contains 48,895 records and 16 columns, representing Airbnb listings across the five boroughs of New York City. The dataset appears to be a snapshot from 2019 but contains historical booking information dating back to 2011.

1.2 Column Descriptions

The dataset contains the following fields:

Column	Data Type	Description
id	Integer	Unique identifier for each Airbnb listing
name	String	Title/name of the listing provided by the host
host_id	Integer	Unique identifier for each host
host_name	String	Name of the property host
neighbourhood_group	Categorical	Borough classification (Manhattan, Brooklyn, Queens, Bronx, Staten Island)
neighbourhood	Categorical	Specific neighborhood within the borough
latitude	Float	Geographic latitude coordinate of the property
longitude	Float	Geographic longitude coordinate of the property
room_type	Categorical	Type of space offered (Entire home/apt, Private room, Shared room)
price	Integer	Price per night in USD
minimum_nights	Integer	Minimum nights required for booking
number_of_reviews	Integer	Total number of reviews received
last_review	Date	Date of the most recent review
reviews_per_month	Float	Average number of reviews per month

Column	Data Type	Description
calculated_host_listings_count	Integer	Number of listings the host has in total
availability_365	Integer	Number of days the listing is available in a year

1.3 Data Distribution

- **Geographical Distribution:** The dataset covers all five NYC boroughs with varying densities
- **Temporal Coverage:** Review dates span from March 28, 2011, to July 8, 2019 (approximately 8.28 years)
- **Price Range:** \$10 to \$10,000 per night, with a mean of \$152.72
- **Room Types:** Entire home/apt, Private room, and Shared room categories

2. Data Quality Assessment

2.1 Missing Values

The dataset contains missing values in several columns: - name: 12 missing values - host_name: 13 missing values - last_review: 2,456 missing values (16.5%) - reviews_per_month: 2,456 missing values (16.5%) - Other columns (neighbourhood_group, neighbourhood, latitude, longitude, room_type, price, minimum_nights, number_of_reviews, calculated_host_listings_count, availability_365): 1 missing value each

The missing values in last_review and reviews_per_month likely represent listings that have not yet received any reviews.

2.2 Duplicate Records

No duplicate records were detected in the dataset.

2.3 Outliers and Anomalies

- **Price:** Contains significant outliers, with prices ranging from \$10 to \$10,000. The 75th percentile is only \$176, indicating that prices above this might need careful consideration.
- **Minimum Nights:** Most listings require 1-5 nights minimum stay, but the maximum is 1,250 nights (over 3 years), which seems unusually high.
- **Number of Reviews:** The maximum number of reviews is 629, with a mean of 23.4, suggesting a heavily right-skewed distribution.
- **Reviews per Month:** Values range from 0.01 to 14.58, with the higher values potentially representing extremely popular listings.

2.4 Data Consistency

The data appears generally consistent across fields, with appropriate values in each column. Geographical coordinates match the expected range for New York City.

3. Relevance for Project Requirements

3.1 Coverage and Scope

- **Time Coverage:** The dataset meets the 5+ year requirement with 8.28 years of review history (2011-2019).
- **Volume:** With nearly 50,000 records, the dataset provides ample data for robust analysis and modeling.
- **Variables:** The dataset contains a rich mix of categorical, numerical, and geographical variables suitable for various data mining techniques.

3.2 Suitability for CRISP-DM Methodology

The dataset is highly suitable for the CRISP-DM methodology: - **Business Understanding:** Can address various business questions related to Airbnb pricing, booking patterns, and property characteristics. - **Data Understanding:** Contains diverse variables for exploratory analysis. - **Data Preparation:** Requires cleaning for missing values and outlier treatment. - **Modeling:** Supports multiple modeling approaches: - **Classification:** Predicting room type or price categories - **Regression:** Modeling pricing or review frequencies - **Clustering:** Identifying similar neighborhoods or listing types - **Time Series:** Analyzing booking trends over the years

3.3 Potential Project Questions

The dataset can address several interesting questions: 1. What factors most influence Airbnb pricing in New York City? 2. How do Airbnb listings differ across the five boroughs? 3. Is there a relationship between listing characteristics and review frequency? 4. Can we identify clusters of similar listings or neighborhoods? 5. How have Airbnb trends changed in NYC from 2011 to 2019?

4. Conclusion

The New York City Airbnb dataset is well-structured, reasonably clean, and highly relevant for the data mining project requirements. It offers sufficient temporal coverage (over 8 years) and contains a rich set of variables suitable for exploratory analysis and model building. The dataset requires some preprocessing to handle missing values and outliers but overall provides an excellent foundation for applying the CRISP-DM methodology and developing multiple data mining models.